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Comparing the effects of ChatGPT and automated writing evaluation on students' writing and ideal L2 writing self

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ABSTRACT

The affordances of ChatGPT in language learning and teaching have gained increasing traction. While studies began to investigate the potential of ChatGPT as a feedback provider, little attention was given to ChatGPT's potential impact on students' writing performance and the ideal L2 writing self vis-à-vis the established automated writing evaluation systems (AWE). To address these gaps, a sequential explanatory mixed methods design was adopted. One hundred and fifty second-year university students from three writing classes in a Chinese public university were recruited and randomly divided into a ChatGPT group, an AWE group, and a control group. After an eleven-week intervention, the ANCOVA results showed that while the ChatGPT group scored significantly higher than the AWE group and the control group in post-writing performance as measured by their writing score, in terms of students' ideal L2 writing self, the ChatGPT group performed significantly lower than the AWE group with a medium effect size. Qualitative analysis of students' reflection papers revealed students' (over)reliance on the tool and the accompanying loss of creativity and agency. Pedagogical implications as well as directions for future research are also discussed.

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ChatGPT; automated writing evaluation (AWE) systems; ideal L2 writing self; feedback

Introduction

Advances in technology have been transforming language learning and teaching, especially for writing instruction (Barrot, 2021; Godwin-Jones, 2022, 2024). The release of ChatGPT in November 2022 may further transform writing instruction. ChatGPT is a form of generative artificial intelligence (GenAI) based on large language models. It can interact naturally with humans in a human-like manner and it has been assessed to

possess good knowledge in most subject matter (Tian et al., accepted). Its affordances have spurred a line of research into its capabilities and potential applications to aid writing as a form of automated feedback (Bucol & Sangkawong, 2024; Guo & Wang, 2023; Pfau et al., 2023; Steiss et al., 2024; Su et al., 2023).

In writing research, ChatGPT tended to be envisioned as an upgraded assessment tool, in comparison to the well-established automated writing evaluation (AWE) systems (Bucol & Sangkawong, 2024; Godwin-Jones, 2024; Kohnke, 2024; Su et al., 2023). AWE systems, such as Grammarly, Criterion, and Pigai, were designed to provide both summative scores and formative feedback for learners' writing, based on natural language processing, artificial intelligence, latent semantic analysis, etc. (Barrot, 2021; Shi & Aryadoust, 2023, 2024; Warschauer & Ware, 2006). The studies of AWE in the past two decades have shown that AWE has a positive overall effect on writing quality and is helpful for L2 students to improve their linguistic accuracy (Ding & Zou, 2023; Karatay & Karatay, 2024; Shi & Aryadoust, 2024; Zhai & Ma, 2023). The AWEs were narrow AI as they employed less complex algorithms that are designed for particular tasks such as grammar checking (Schmidt & Strasser, 2022), while ChatGPT, based on large language models (LLMs), is able to check grammar, sentences and provide feedback based on rubrics, it is also able to produce complete essays when given a prompt (Escalante et al., 2023), and is believed to afford more interactive and customized learning (Godwin-Jones, 2024; Kohnke, 2024; Wang et al., 2024).

As such, it appears that ChatGPT is a perfect replacement for AWE, but research on ChatGPT indicated that this is not always the case. Karataş et al. (2024) found that learners held mixed attitudes toward using ChatGPT, and Niloy et al. (2023) found that using ChatGPT exerted a negative impact on students' writing. According to sociocultural theory (Vygotsky, 1978), technologies such as ChatGPT and AWEs are regarded as artifacts that bring about varied forms of social interaction. The feedback that forms scaffolding for learning, be it from human beings or these tools, would be most effective when it is given at a level at which the learner can actively engage with it and learn from it. Therefore, it is worthwhile and informative to compare the effects of ChatGPT and AWE on learners' writing.

Moreover, these tools have impacted not only the quality of writing but also learners-related variables, with motivation being one of the most significant. Quite a few studies have started examining the effect of AWE and ChatGPT on motivation and found positive impacts (Huang & Mizumoto, 2024; Song & Song, 2023; Wilson & Czik, 2016). However, as stated in Stockwell and Wang (2024a, 2024b), although often regarded as means of motivating learners, technology itself does not necessarily lead

to increased motivation. Thus, it is necessary to continuously examine the effect of ChatGPT on learners' motivation. Ideal L2 writing self, as a strong predictor of writing achievement and motivation (Tahmouresi & Papi, 2021), fits well as an important variable in our present study.

The role of ChatGPT in writing will be better understood by comparing its impact on students' writing performance and ideal L2 writing self in writing against the more established AWE systems, which form the rationale of the present study (Ding & Zou, 2023; Shi & Aryadoust, 2024).

Literature review

AWE feedback and ChatGPT feedback

Feedback plays an important role in enhancing learners' writing ability (Hattie & Timperley, 2007; Hyland & Hyland, 2019). While feedback conventionally comes from teachers or peers, the advancement of technology has led AWE systems to be widely used and researched as a feedback provider (Ding & Zou, 2023; Karatay & Karatay, 2024; Shi & Aryadoust, 2024; Warschauer & Ware, 2006). Research on AWE centers on AWE feedback accuracy as compared to human feedback, perceptions towards AWE feedback, and the effects of AWE feedback (for a detailed review, see Ding & Zou, 2023; Shi & Aryadoust, 2024; Stevenson & Phakiti, 2014). The studies of feedback provided by ChatGPT have followed a similar research route on examining the accuracy of ChatGPT feedback as well as its effects (Barrot, 2023a; Su et al., 2023; Wang et al., 2024).

In terms of investigating the accuracy of AWE feedback and ChatGPT feedback, the following two statistical concepts were often used: precision (the ratio of the number of correct feedback by the number of all reported feedback) and recall (the proportion of errors that have been detected in relation to all errors) (Leacock et al., 2010). Studies of AWE feedback found that hardly any existing AWE systems met the stringent criterion of 0.9 (Burstein et al., 2003) except for certain types of errors (e.g. spelling errors, article errors) (Ranalli & Yamashita, 2022). On the other hand, the current research on ChatGPT feedback found that ChatGPT produced a score of 91.8% for precision and 63.2% for recall for feedback on students' argumentative writing (Wang et al., 2024), and a high precision rate (99%) for identifying language errors on L2 learners' essays (Pfau et al., 2023).

Both AWE feedback and ChatGPT feedback were compared to human feedback, and both have their strengths and weaknesses (Guo & Wang, 2023; Lu et al., 2024; Steiss et al., 2024; Wang et al., 2024; Zhang & Hyland, 2018). Research showed that human feedback addressed more error categories while AWE feedback tended to highlight errors in

students' writing (Zhang & Hyland, 2018). Regarding ChatGPT feedback, Guo and Wang (2023) found that ChatGPT produced a much larger amount of feedback distributed evenly on content, organization, and language while teacher feedback mainly focused on content and language of students' argumentative writing. Similarly, Wang et al. (2024) revealed that while feedback by ChatGPT was comprehensive and textual-based, its emotion-related feedback was only limited to the linguistic level; on the other hand, teacher feedback took the teacher's personal teaching experience and humanistic affect into consideration when providing feedback. Overall, it seems that human raters were able to provide feedback of higher quality compared to ChatGPT (Steiss et al., 2024). A recent study done by Kohnke (2024) examined university students' perception towards AWE and GenAI tools and found that students favored the letter as more comprehensive and authoritative.

The effects of AWE feedback and ChatGPT feedback were the most frequently researched area. The effects of AWE on students' writing have been reported to be overall positive (Zhai & Ma, 2023), but the results are not always consistent as Mohsen's (2022) meta-analytic study revealed only a medium overall effect when comparing AWE to non-AWE conditions. In a similar vein, studies of ChatGPT on writing found that students assisted by ChatGPT achieved significantly higher learning outcomes (i.e. organizational skills, coherence, grammar, and vocabulary (Boudouaia et al., 2024; Song & Song, 2023)). The results, though, are not always positive, either: Escalante et al. (2023) found that in terms of learning outcomes, there was no significant difference between the group receiving ChatGPT-generated feedback and the group receiving tutor feedback; the use of ChatGPT also raised concerns including learning loss, losing authorial voice, producing unintelligent texts, and violating academic integrity, (Zou & Huang, 2023); Niloy et al. (2023) found that ChatGPT had an adverse impact as students using ChatGPT scored significantly lower in all four aspects of creative writing—content accuracy, content presentability, elaboration and similarity.

While sharing so many similarities, ChatGPT seems to be taken for granted as a replacement for the traditional AWE systems though without any empirical evidence. This necessitates the present study to investigate and compare their effects on learners' writing gains.

The ideal L2 writing self

The ideal self refers to an individual's idealized version of himself/herself in the future (Dörnyei, 2005; Markus & Nurius, 1986). In language learning, the ideal L2 self is a key component of the L2 motivational self system (Dörnyei, 2009), and research has shown that the ideal self led to

an eager inclination for maximal L2 use and higher levels of L2 achievement (Papi & Khajavy, 2021). Recent studies have applied the ideal L2 self to L2 writing and found that this concept was a strong predictor of writing achievement and motivation (Tahmouresi & Papi, 2021). The ideal L2 writing self represents the ideal image of an L2 writer that a learner wants to be in the future. The distance between learners' current level and their desired future selves creates feelings of discomfort and serves as a motivating factor for students to strive toward their desired selves (Tahmouresi & Papi, 2021). Furthermore, studies have shown that compared to the general ideal L2 self measure, the ideal L2 writing self was an important contributor to L2 writing achievement (Tahmouresi & Papi, 2021; Zhao et al., 2023).

The ideal L2 self is not just a 'snapshot in time' of learners' motivation (Aubrey & Philpott, 2021, p. 441). Although fewer studies looked at how the L2 ideal self changes as a result of an intervention, it could also be positively affected with appropriate assistance (Kim, 2009). For instance, a thoughtful homestay owner proficient in the target language can create an ideal L2 learning environment for homestay L2 learners by providing appropriate feedback during daily conversations using the target language. The L2 learner's perception of their ideal L2 self might also be positively affected (Kim, 2009).

Studies on the impact of AWE and ChatGPT on language learners' motivation have been reported to produce positive results (Song & Song, 2023; Wilson & Czik, 2016), though not using the construct of ideal L2 self. Nevertheless, a more recent study done by Huang and Mizumoto (2024) investigated the impact of ChatGPT on students' motivation from the lens of ideal L2 self and found enhanced motivation in participants. However, the more specific ideal L2 writing self, which is more closely related to the L2 writing context as proposed in Tahmouresi and Papi (2021), has not yet been investigated.

Given the important roles that the ideal L2 self plays in motivating language learners and the potentially positive impact of AWE and ChatGPT on learners' motivation and other psychological traits, it is necessary to examine the role of AWE and ChatGPT in influencing learners' ideal L2 writing selves. Accordingly, we hypothesize that feedback by ChatGPT and AWE could positively affect learners' ideal L2 writing self.

Theoretical framework

The present study is grounded in sociocultural theory (SCT) (Vygotsky, 1978). There are three closely related key components in this theory: mediation, Zone of Proximal Development (ZPD), and scaffolding. Mediation is considered a dynamic interaction between less capable

learners and more capable others, in which the more capable ones help the less capable ones improve (Rathert & Cabaroğlu et al., 2024). Feedback, in this sense, could be regarded as one kind of mediation to support a learner's learning. ZPD refers to 'the distance between the actual developmental level as determined by independent problem solving and the level of potential development as determined through problem solving under adult guidance or in collaboration with more capable peers' (Vygotsky, 1978, p. 86). The concept of ZPD emphasizes the critical role that assistance plays in promoting development. This kind of assistance, provided by a more capable peer or expert, is referred to as scaffolding.

In terms of AWE feedback, Xu and Zhang (2022) examined the impact of Pigai (an AWE system) on students' writing and found that AWE feedback on linguistic errors was within ZPD of students with lower language level while below high-level students' ZPD, and that AWE failed to provide feedback within high-level students' ZPD. Regarding ChatGPT feedback, as mentioned above, studies have found that ChatGPT was able to offer large amounts of feedback on content, organization, and language (Guo & Wang, 2023), and the real-time feedback by ChatGPT was akin to a more knowledgeable virtual peer (Song & Song, 2023), and it might be assumed that ChatGPT was better than traditional AWE when scaffolding learners. However, as argued by Lee (2019), unfocused and comprehensive feedback might not be as effective as focused feedback because focused feedback from teachers could help students make better progress within their ZPD. Thus, from the SCT perspective, it is meaningful and informative to compare these two types of feedback on students' learning gains.

Research questions

Although there are great similarities between AWE and ChatGPT feedback, to our knowledge, no studies have been done to compare their effects on students' writing and the ideal L2 writing self. Comparing ChatGPT against the more established AWE systems will be informative and shed light on future practice and research, given the inconsistent results of using ChatGPT in teaching and learning writing and the mixed perceptions towards ChatGPT (Niloy et al., 2023; Zou & Huang, 2023), together with the continuing need for deeper understanding of both AWE systems and ChatGPT (Ding & Zou, 2023; Shi & Aryadoust, 2024). Thus, our research questions are:

1. Are there differences in students' writing performance after using ChatGPT and AWE for an eleven-week intervention?

2. Are there differences in the change of students' ideal L2 writing self after using ChatGPT and AWE?
3. If the differences exist, how do students' perceptions of ChatGPT based on their use contribute to such differences?

Research design

To answer our research questions, we adopted the sequential explanatory design of mixed methods as suggested by Creswell and Creswell (2023). We first conducted the quantitative procedure, employing a pre-test and post-test between-subject design, followed by a qualitative paradigm, in which data was collected through participants' reflection papers and analyzed qualitatively. Adopting such an explanatory design enabled us to first present the quantitative findings, and give further explanations based on the qualitative results.

Context and participants

This research took place in a College English Writing course at a public university in Eastern China. The course, typically offered in the third and fourth semesters, is designed to improve the written communication skills of non-English majors with a focus on language accuracy and variety, paragraph and essay structure, and coherence, as well as prepare students for the writing tasks in high-stakes tests. The primary coursebook for this course is 'English Writing: From Sentence to Paragraph to Theme' by Shi and Shuai (2010). The class size for this course is usually around 50 students. To mitigate the potential impact of confounding variables, three out of five writing classes, all taught by the same instructor, were randomly selected for this study. Prior to the experiment, we conducted a briefing session with the instructor about our research purpose, research design, and research procedure, and the instructor agreed to participate in our experiment. Upon the commencement of the intervention, all students were informed of the confidentiality measures in the experiments and the right to withdraw from the experiment at any time without consequence. In total, 150 participants from three writing classes agreed to take part in the research and were randomly assigned to a ChatGPT group ($n=50$), an AWE group ($n=50$), and a control group ($n=50$). They are second-year non-English major students coming from disciplines including engineering, science, and arts. Their ages ranged from 18 to 22 years old, with the mean being 20.11 years old. Their language proficiency level was low-intermediate to intermediate, as measured through self-reported

College English Test Band Four (CET-4) score. At the end of the experiment, the complete data of 118 students were collected. Further screening removed 10 participants who completed the questionnaire within less than 100s. The final dataset consisted of data from 35 participants (mean age = 20.29, $SD=0.71$; 48.57% female) for the ChatGPT group, 42 participants (mean age = 20.00, $SD=0.88$; 59.52% female) for the AWE groups, and 31 (mean age = 20.07, $SD=0.83$; 45.16% female) for the control group.

Pedagogical materials

For AWE systems, this study used Pigai as the main system as it is predominantly used in China for English writing practice and evaluation (Ding & Zou, 2023), and has been the most frequently examined AWE system in empirical research with satisfactory efficacy (Shi & Aryadoust, 2024). In addition, we also introduced students Youdao writing (<https://write.youdao.com/#/homepage>), another AWE system that could revise and improve students' writing at the sentence level. All features of Youdao writing were free of charge and were publicly accessed throughout the whole study.

For ChatGPT (OpenAI, 2023), we provided students with the following exemplar prompts during the training session based on Kohnke et al. (2023) and Guo and Wang (2023) and encourage students to use all of them for every revision task: (1) You are an expert writer, please point out and correct all the language mistakes in the following essay (for all language errors); (2) You are an expert writer, please revise and improve the following paragraph(s)/essay (for local/global language improvement); (3) You are an expert writer, please reorganize the following paragraph(s)/essay to make it more coherent (for local/global organization); (4) You are an expert writer, please make the following argument more detailed (for local content improvement within a paragraph); (5) You are an expert writer, please provide a detailed example for the following argument (for local content improvement within a paragraph).

Instruments

Data were collected using a questionnaire that included a total of 58 items. All items were rated on a 5-point Likert scale ranging from 1 (totally disagree) to 5 (totally agree). The present study was part of a larger project that included other constructs. Therefore, only the data related to the present study are reported.

Ideal L2 writing self

The scale used for measuring the ideal L2 writing self consisted of six items (see [Appendix A](#)) adapted from the questionnaire validated by Tahmouresi and Papi (2021). Example items include ‘I imagine a day where I am easily writing emails in English.’ The Cronbach α was 0.91 in the pre-test, and 0.96 in the post-test in our study, demonstrating satisfactory reliability.

Writing prompts

To ensure the reliability of the writing prompts, the genre of all four writing prompts employed in this study was argumentative and was adapted from a pool of writing sections in College English Test Band Four (See [Appendix B](#)).

Reflection paper

Immediately after the post-test, participants in the AWE group and the ChatGPT group were asked to write a reflection paper based on their experience of using such feedback, discussing what positive and negative impact using the tools had on their own writing, and whether they would consider using these tools to assist their writing and why. To facilitate students’ thoughts, they were free to choose the language they felt comfortable with to write the reflection paper, and most students completed this task in Chinese.

Procedures of the study

The whole procedure for this study is presented in [Figure 1](#). The experiment started in week one and lasted for eleven weeks. Prior to the intervention, in week one, the same pre-test writing prompt and questionnaires were administered to all three groups to investigate their baseline writing performance and ideal L2 writing self. In week two, as part of the course curriculum, students in all groups were asked to form a group of two or three and give peer feedback on each other’s first writing in class. As the length of participants’ essays typically was around 150 words, students were given around 20 min in class for giving peer feedback. For the AWE and ChatGPT groups, a brief training session was also given in week two; students were shown how to submit their writing and receive feedback using Youdao in the AWE group ([Figure 2](#)) and how to use ChatGPT assisted with examples from Su et al. (2023) and Kohnke et al. (2023) in the ChatGPT group ([Figure 3](#)). After class, participants of all groups were asked to revise their first writing based on the peer

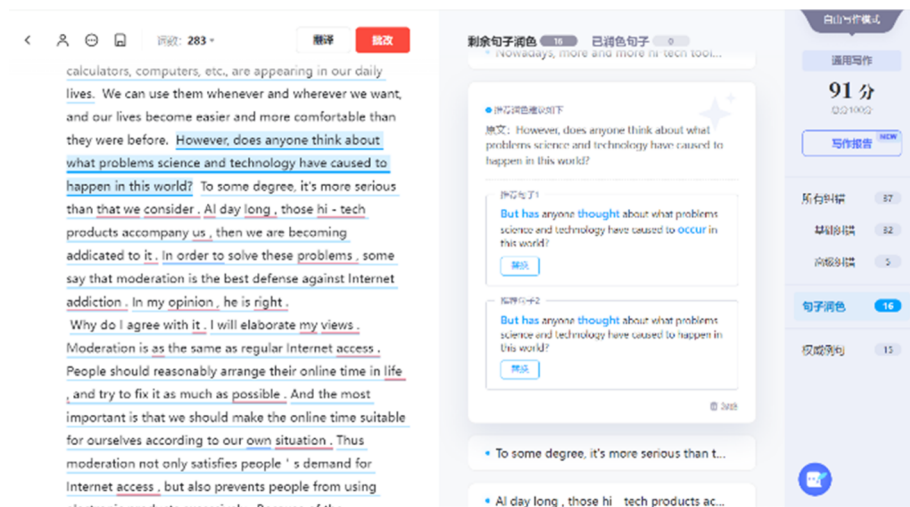


Figure 1. Procedures of the study.

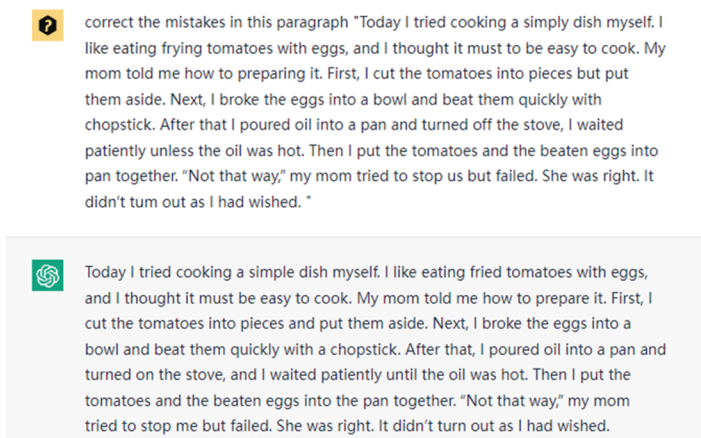


Figure 2. A screenshot of feedback by Youdao Write.

feedback they received; meanwhile, the AWE group could use AWE systems to assist with their revision, while the ChatGPT group could revise with assistance offered by ChatGPT. To ensure participants' active engagement with ChatGPT, the instructor encouraged participants to use exemplary prompts to generate feedback for revision. The same procedure was repeated for prompt two and prompt three; the only difference was that the instructor assessment was given on writing of prompt two for all three groups as required in the course curriculum. No assignment was arranged for Week Six due to a holiday. In week eleven, the post-test of the same questionnaire and a writing prompt were conducted among all three groups in class. The in-class writing for the post-test (prompt four) comprised 15% of the final course grade, while each of the other three

ANCOVA is regarded as the appropriate analytic method when testing for group differences in change from the pre-test to the post-test. Through controlling the effects of one or more continuous variables known as covariates (in our case, the pre-test score) that might influence the dependent variables, ANCOVA helps reduce error variance and eliminate confounds, and by doing so, increase the statistical power and precision (Field, 2018).

The independent variable was the type of intervention, i.e. the control group, the AWE group, and the ChatGPT group. The pre-test data was controlled as a covariate and the dependent variables were the corresponding post-test data including students' writing scores and their responses to the questionnaire of the ideal L2 writing self.

To answer the last research question, we used qualitative analysis methods following Saldaña (2016) to analyze the reflection papers written by the experiment groups. Two coding methods were adopted: open coding and focused coding. Two researchers coded the data independently using open coding and focused coding in the first two rounds. Then two researchers discussed together the disagreements in codes, with reference to studies by Zou and Huang (2023) and Karataş et al. (2024), and resolved the disagreements. The whole analysis was iterative until the themes were finalized.

Results

Quantitative results

The data of the pre-test scores for the ideal L2 writing self and the writing score of prompt one of the three groups was presented in Table 1. Although the mean score and the standard deviation for each group across the three variables were slightly different, ANOVA analysis showed that there was no statistically significant difference among the three groups in both variables.

Changes in writing performance

Table 2 presents the results of the post-test for writing performance. It showed that compared to the mean score of the control group 9.29

Table 1. One-way ANOVA results of pre-tests.

	Group	N	M	SD	SE	F	p
Ideal L2 writing self	ChatGPT	35	4.06	0.70	0.12	2.20	0.12
	AWE	42	4.17	0.58	0.09		
	Control	31	3.84	0.74	0.13		
Writing score	ChatGPT	35	8.80	1.59	0.27	1.79	0.17
	AWE	42	8.90	1.53	0.24		
	Control	31	8.26	1.39	0.25		

Table 2. ANCOVA results of post-experiment writing performance for three groups.

Group	Post-test			Univariate ANCOVA				
	N	M	SD	Adjusted Mean	SE	F	p	η^2
ChatGPT	35	10.51	1.43	10.50	0.23	5.87	0.00	0.10
AWE	42	9.81	1.22	9.74	0.21			
Control	31	9.29	1.59	9.41	0.24			

Table 3. Pairwise comparison of post-test writing performance for three groups.

(I) groups	(J) groups	Mean	SE	Sig.	95% C.I. for Difference		d
		Difference (I-J)					
ChatGPT	AWE	.763*	0.31	0.04	0.02	1.51	0.58
ChatGPT	Control	1.089*	0.33	0.00	0.28	1.90	0.72
AWE	Control	0.33	0.32	0.94	-0.46	1.11	0.24

*The mean difference is significant at the .05 level.

Table 4. ANCOVA results of post-experiment Ideal L2 writing self for three groups.

Group	Post-test			Univariate ANCOVA				
	N	M	SD	Adjusted Mean	SE	F	p	η^2
ChatGPT	35	3.71	0.85	3.70	0.10	5.01	0.01	0.09
AWE	42	4.23	0.64	4.14	0.10			
Control	31	3.75	0.82	3.89	0.11			

($SD=1.59$), the mean writing score of both the AWE group and ChatGPT group were higher: 9.81 ($SD=1.22$) and 10.51 ($SD=1.43$) respectively. Assumptions including normality, homogeneity of variance, the independence of the covariate and treatment effect, and homogeneity of regression slopes were checked and met before running ANCOVA. One-way ANCOVA was run with participants' pre-test writing score as the covariate, and the result demonstrated a significant difference with a medium effect size among the post-test results of the three groups ($F(2, 104) = 5.87$, $p = .00$, partial $\eta^2 = .10$).

To find out the specific differences, the mean writing scores of all groups were contrasted using the post hoc Bonferroni test (Table 3). The results showed that the mean writing scores of the ChatGPT group were significantly higher than the mean scores of the control group ($p = .00$, 95% C.I. = [0.28, 1.90], $d=0.72$), as well as from the AWE group ($p = .04$, 95% C.I. = [0.02, 1.51], $d=0.58$), with a medium effect size. There was no statistically significant difference between the AWE group and the control group ($p = .94$).

Changes in the ideal L2 writing self

Statistics of the post-test for Ideal L2 writing self were presented in Table 4. Descriptive statistics showed that the AWE group had the highest mean

score (4.23, $SD=0.64$) on ideal L2 writing self. One-way ANCOVA was run with participants' pre-test ideal L2 writing self as the covariate, and the result showed a significant difference ($F(2, 104) = 5.01$, $p = .01$, partial $\eta^2 = .09$), demonstrating there was a significant difference with a medium effect size among the post-test results of the three groups.

Similarly, we compared the mean ideal L2 writing self scores of all groups using the post hoc Bonferroni test (Table 5). The results showed that the mean ideal L2 writing self scores of the ChatGPT group were significantly lower than the mean scores of the AWE group ($p = .01$, 95% C.I. = $[-0.79, -0.10]$, $d=0.59$), with a medium effect size. No statistically significant difference was found between the AWE group and the control group ($p = .26$) or between the ChatGPT group and the control group ($p = .68$).

Qualitative results

To answer the third research question, we approached the collected reflection paper (22 papers from the ChatGPT group and 27 papers from the AWE group) with qualitative analysis, coded the data, and organized the themes revolving around the first two research questions. We underwent three rounds of iterative coding before identifying all the themes. A summary of the themes and their percentage with respect to the total number of reflection papers in each group was presented separately in Figures 4 and 5, corresponding to the first and the second research questions.

Themes related to writing performance

The qualitative analysis revealed that students from both groups exhibited similar patterns, albeit with some differences. In the ChatGPT group, after being familiar with the features of ChatGPT, participants expressed that they were impressed by its powerful capability of generating ideas (54.5%) and by the advanced version of their own writing after revision as there were more advanced vocabulary and sentences (40.9%). One respondent (S4, chemistry engineering) noted that ChatGPT was comparable to a teacher as *'it can revise the writing from the deeper level (content) and add its own thing... it did not overuse common transitions, and*

Table 5. Pairwise comparison of post-test Ideal L2 writing self for three groups.

(I) groups	(J) groups	Mean	SE	Sig.	95% C.I. for Difference		d
		Difference (I-J)					
ChatGPT	AWE	-0.443*	0.14	0.01	-0.79	-0.10	0.59
ChatGPT	Control	-0.19	0.15	0.68	-0.56	0.19	0.23
AWE	Control	0.26	0.15	0.26	-0.10	0.62	-0.34

*The mean difference is significant at the .05 level.

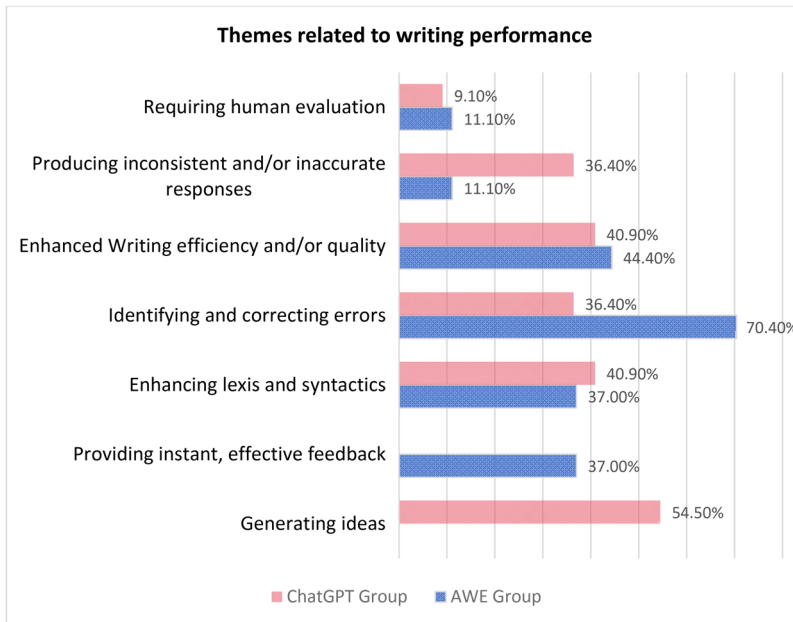


Figure 4. Themes related to writing performance.

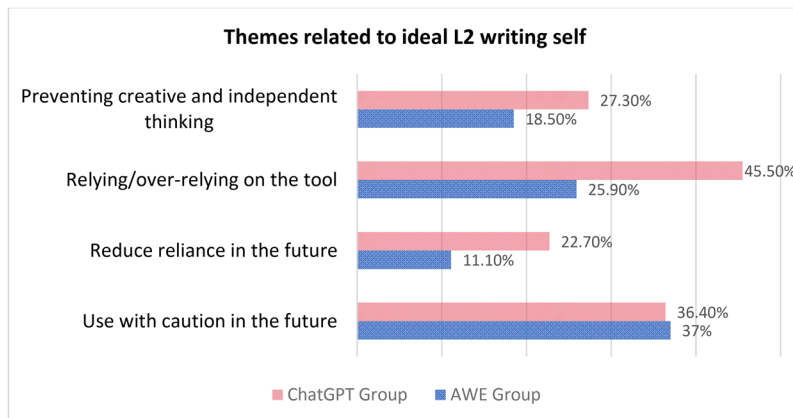


Figure 5. Themes related to ideal L2 writing self.

it revised my essay to make it more coherent by using advanced cohesion techniques (e.g. another). This affordance of ChatGPT was more powerful than what AWE was capable of. By correcting errors, and providing more choices both in lexical, syntactical, and textual levels, ChatGPT greatly facilitated participants' writing process and improved their writing quality. Furthermore, feedback by ChatGPT went beyond surface-level corrections (grammar, vocabulary, spelling) and was capable of revising students' essays at a deeper level (content). These positive impacts and the facilitating process are vividly illustrated in the following excerpt:

ChatGPT can quickly generate textual content, reducing the time and effort costs involved in my writing process. It can also improve my language style. ChatGPT provides me with high-quality and grammatically right language templates, helping me improve my language style and express thoughts and ideas more accurately. Additionally, it assists me in expanding my vocabulary. Through the output of the GPT-3.5 model, I can encounter a greater variety of English words and phrases, which aids in expanding my vocabulary and expressive abilities. Most importantly, it helps me enhance my writing skills. The GPT-3.5 model can automatically handle tasks such as grammar, spelling, and punctuation corrections in my writing, thereby assisting me in improving my writing proficiency (S12, automotive engineering).

By comparison, the AWE group highlighted the features of AWE systems: providing instant effective feedback and correcting language errors. Similarly, quite a few participants expressed that their writing improved using AWE. One respondent noted that *'Pigai could help me quickly identify problems and shortcomings in my own writing, and I could quickly receive suggestions on how to correct them, thus making my writing more professional'* (S55, software engineering)

However, similar to AWEs, ChatGPT is not without limitations: participants reported some weaknesses of ChatGPT based on their using experience: the inconsistent/inaccurate feedback (36.40%), and the need for human evaluation (9.10%). For instance, one participant (S17, computer science) complained that *'I gave the same prompt to ChatGPT but got different responses each time, although there might be a high level of similarity among all the responses.'* Inaccuracy was reported when students found that the prompts they formulated failed to produce the desired response from ChatGPT. The inconsistency in responses further consolidated students' concern about the issue of ChatGPT. It was noted that this issue was not limited to ChatGPT; Students in the AWE group also reported that they encountered the inaccuracy issue, albeit to a lesser degree. For example, one respondent stated that *'some suspected errors by Pigai are incorrectly identified, and some recommended expressions don't quite match our intended meaning or are somewhat unnecessary. There is still room for improvement in these areas.'*

Themes related to the ideal L2 writing self

The reason why the ChatGPT group had lower scores compared to the AWE group in terms of the ideal L2 writing self could be partly due to reliance/overreliance on the tools (45.50%), and loss of creativity and learner agency (27.30%) as revealed in our qualitative data. As noted by one participant (S7, chemistry engineering), *'Over-reliance on ChatGPT*

made me not willing to write any essay by myself, and I just want to enjoy 'the ready-made dinner', and I could not think on my own.' The concern of 'agency loss' was echoed by a number of participants who stated that using ChatGPT prevented creative and independent thinking. An extreme case was reported by one participant (S14, nuclear engineering) who wrote that *'gradually and unconsciously, I began to 'copy and paste' the writing generated by ChatGPT'*. Though not many, still two participants wrote that ChatGPT affected their writing improvement, as one (S20, computer science) wrote *"ChatGPT drove us to pursue perfection by offering accurate suggestions, and we tend to spend more time on improving the text instead of the 'true writing'"*. Similar to the first theme, there was a similar observation in the AWE group: students using AWE feedback also expressed reliance issue and loss of creative and independent thinking, but to a lesser degree. One respondent wrote *'Using these tools (Youdao and Pigai) helped me improve the elegance and readability of my sentences. However, frequent use also led to laziness and over-reliance on these AI-assisted writing modes (S67, Chinese)'*

Part of the reason was identified in students' statements of future intention on whether or not to use this tool. Many participants expressed their determination to reduce reliance on the tool (22.70%) and others stated that they would use the tool with caution (36.40%). For example, one participant (S10, chemistry engineering) noted *'In the future, I plan to continue using these tools, but I will be more mindful of how I use them. I will strive to strike a balance between using these tools to improve my writing skills and relying on my own abilities and knowledge. I will also make an effort to enhance my English language skills so that I can rely less on these tools and write with more confidence and fluency.'* Another participant (S18, computer science) noted that *'although I would continue to use this tool in the future, but I will try not to use it unless I have to. I need to reduce my reliance on it'*. Similarly, but to a lesser degree, in the AWE group, participants also expressed their cautious use of the AWE programs in the future.

Overall, compared to the AWE group, the ChatGPT group showed a higher percentage of concern over the dominating role of ChatGPT in their writing as ChatGPT led them to be more reliant, and affected their learner agency by preventing independence and creativity. As such, participants were more cautious about the use of ChatGPT in the future and expressed their intention to reduce reliance on this tool.

Discussion

With the aim of better understanding the effects of ChatGPT and students' using experience, this research adopted the sequential explanatory

design of mixed methods to explore and compare the effects of AWE and ChatGPT on students' writing performance, ideal L2 writing self, and their using experience. Our results showed that after the eleven-week intervention, the ChatGPT group scored significantly higher than the control group and the AWE group on the post-test writing performance with a medium effect size; for the ideal L2 writing self, the ChatGPT group scored significantly lower than the AWE group with a medium effect size, and no statistically significant difference was found between the ChatGPT group and the control group. Qualitative analysis of students' reflection papers revealed that participants were fully aware of the affordances by ChatGPT on the one hand, and on the other hand, they were also concerned with their reliance on the tool and the accompanying loss of creativity and agency and expressed their reserved attitude toward future intention to use the tool.

Compared to conventional AWE systems such as Pigai, ChatGPT was conjectured to offer more affordances such as providing feedback and enhancement on multiple aspects of writing (content, organization, vocabulary, and grammar) (Barrot, 2023a, 2023b; Ding & Zou, 2023; Wang et al., 2024). The qualitative results in our study suggested that participants were well-informed of the powerful capabilities of this tool. With these affordances, more opportunities to learn and improve language skills were created, and students had more ways to engage with the ChatGPT feedback; therefore, it could support students' writing with more learning resources and opportunities. Our eleven-week intervention showed that the ChatGPT group performed better than the control group as well as the AWE group, echoing the findings in Song and Song (2023). Hence, it seems true that ChatGPT could enhance writing performances, and it could support university EFL students' writing.

As for the ideal L2 writing self, we found that there was no statistically significant difference between the score of the ideal L2 writing self for the ChatGPT group and the control group, but compared to the AWE group, the ChatGPT group scored significantly lower of the ideal L2 writing self. Though surprising, the reasons might be traced in the results of the analysis of the reflective paper as mentioned earlier. Participants in the ChatGPT group stated their (over) reliance on the tool, which in turn prevented their creative and/or independent thinking, thereby affecting their learner agency. This finding is in accordance with the statement that technology itself does not create autonomy (Stockwell & Wang, 2024a, 2024b). They were aware of these concerns and expressed their intention to use it cautiously or reduce reliance in the future. These findings consolidated the concern expressed in Kasneci et al. (2023) that the effortlessly generated information could negatively impact students' critical thinking and problem-solving skills as ChatGPT

simplified the process of acquiring information, and by doing so increased students' laziness and counteract their interest in independent learning, and that without guidance, students tend to rely on instead of learning from AI assistance, which weakens student agency (Darvishi et al., 2024). Gradually, this may lead to 'AI enfeeblement' (Carlson et al., 2023), which happens when humans are completely reliant on AI. Students' reliance and overreliance on the tool led them to believe that they were less creative and less able to think on their own, which widened the gap between their current self and their ideal L2 selves. On the other hand, students were fully aware of and worried about losing their authorship and learner agency as they realized that ChatGPT led them to be more reliant and affected their independence and creativity, as also found in Zou and Huang (2023). Facing this dilemma, participants in our study started to think about resisting ChatGPT by determining to reduce reliance on the tool as well as using the tool with caution in the future in order to sustain their dominating role as a learner and a writer.

Although our study was conducted in an EFL context in China, with participants at a low-intermediate to intermediate language proficiency level, the (over)reliance on technological tools, particularly ChatGPT, as highlighted in our findings, resonates with broader research trends. This reliance is likely to impact students' ideal L2 writing self-concept, not only in ESL or EFL environments but also in the learning of languages other than English. The strong intention expressed by our participants to reduce their reliance in the future indicates their awareness of maintaining their learning agency. This insight aligns with the human-centered approach towards AI use in education as advocated by UNESCO (Miao & Holmes, 2023). Educators as well as policy makers need to take this into account when designing courses or policies.

Hence, it will require educators to guide students to use it adaptively. While training workshops on how to better use ChatGPT for writing would be helpful, an additional effective practice is to encourage students to submit their responses and revisions to ChatGPT for feedback accompanied by brief explanatory notes, which fosters students' critical evaluation of their usage of ChatGPT, thus helping them to retain their agency rather than using the output in a 'copy and paste' manner. Teachers could also consider designing collaborative writing tasks that requires students to collaboratively process feedback provided by ChatGPT as collaborative learning has been conclusively proven to be highly effective in promoting deeper understanding (Mujtaba et al., 2021). These activities will help students understand and use ChatGPT in a responsible and creative way.

Pedagogical implications

Based on the results and discussion, to maximize the positive effects of ChatGPT on writing, and mitigate the negative effects on learners' ideal L2 writing self, we propose the following suggestions. First of all, educators need to refine the learning objectives based on students' ZPD. Considering the participants' language level in our research, the systematic and organized feedback provided by the AWE systems might have a better scaffolding effect for low intermediate to intermediate language level learners. Compared to AWE, ChatGPT has far more affordances (Barrot, 2023a, 2023b; Godwin-Jones, 2024; Tseng & Warschauer, 2023), and is not pre-customized for learners of certain language proficiency. The clear, specific list of learning objectives within students' ZPD could work as a prototype for developing prompts that could help students to learn, rather than completing the task. Meanwhile, prompt engineering has gradually been regarded as an important skill that students need to be equipped with (Miao & Holmes, 2023; Tseng & Warschauer, 2023; Walter, 2024). The refined objectives would serve as a starting point for students to improve their prompt engineering capacity. A good example could be found in the study done by Carlson et al. (2023), which offered a prompt template that included all the writing criteria (objectives) that could be borrowed and learned.

Secondly, there is a need to redefine the objectives of language education in order to maintain the agency of both learners and educators. Focusing on providing fixed types of feedback, AWE does not model a better way of writing that can be directly borrowed at the paragraph level or the text level. Hence, with limited affordances, AWE may support students' writing without compromising students' authorship. In the context of developing authorship or students' voices, it seems that AWE could serve as a more appropriate choice of technology. Even so, the qualitative results in our study showed that students in AWE group also reported (over) reliance, which resonated with previous studies (e.g. Zhang & Hyland, 2018). The more affordances of ChatGPT, in our study, seem to lead students to rely more on ChatGPT. This over-relying on the generating features at the paragraph level and textual level might deprive students' learner agency and affect their ideal L2 writing self. As such, to retain learners' agency and ensure proper use of ChatGPT, educators need to reconceptualize the core objectives of language learning and teaching that will not be easily replaced by ChatGPT; among them, AI literacy including prompt engineering and critical thinking skills are suggested to be incorporated into the objectives to enhance students' awareness of responsible and ethical use of the tool (Godwin-Jones, 2024; Tseng & Warschauer, 2023).

Conclusion

This research employed a sequential explanatory design to explore and compare the effects of ChatGPT and AWE on students' writing performance, ideal L2 writing self, and using experience. The results of our research partly answered the concerns and calls raised by Ding and Zou (2023). Our results showed that ChatGPT helped students perform better in their writing compared to the control group and the AWE group, but compared to the AWE group, ChatGPT significantly lowered students' ideal L2 writing self. Qualitative results shed light on possible causes: while participants were fully aware of the affordances of ChatGPT feedback, they were also concerned with their (over) reliance on the tool and the accompanying loss of creativity and agency and expressed their reserved attitude toward future intention to use ChatGPT. We suggest educators need to refine the teaching objectives based on students' ZPD to provide a starting point for prompt engineering as well as maintain learner agency. We further suggest that educators need to redefine the teaching objectives to maintain both teachers' and students' learner agency.

Our research is not without limitations. First of all, our study was conducted in an EFL environment, with participants' language proficiency levels being low-intermediate to intermediate; future studies are suggested to look into learners of higher proficiency levels or more diverse samples as well as more diverse language contexts. Secondly, it needs to be acknowledged that the dropout rates of participants in the experiment groups might have introduced a systematic bias in the results. Students who withdrew from the study may have had different experiences or outcomes or were less motivated, which we have no way of knowing. The absence of their data might thus have affected the comprehensiveness of our findings and should be considered when interpreting the results. Thirdly, in terms of data collection, we collected only one-time piece of writing on students' reflections; it is preferred that students write reflection papers after each revision using ChatGPT, which would provide richer insights. In addition, the duration of our study was relatively short, and to gain deeper insights into sustained outcomes, future research may consider conducting experiments with longer intervention periods. Finally, since the revisions were primarily conducted after class due to the curriculum, there is a possibility that students in both the AWE group and the control group might have used ChatGPT independently. Future research could adopt stricter measures to control external variables when conducting experiments in order to enhance the generalizability of the findings.

Disclosure statement

The authors report there are no competing interests to declare.

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Appendix A. The ideal L2 writing self

I imagine a day that I am easily writing emails in English.

我想象未来有一天, 我在轻松地用英语写邮件。

I can see a day that I can write in English fluently and accurately.

我能预见未来有一天, 我可以流畅又准确地用英语写作。

I can see a day that I am posting materials in English on social networks.

我能预见未来有一天, 我在社交网络上用英语发布内容。

I can see a day that I can easily write down my thoughts and ideas in English.

我能预见未来有一天, 我可以轻松地用英语写下我的想法和创意。

I can see a day that I can easily fill out any form in English.

我能预见未来有一天, 我可以轻松地用英语填写任何表格。

I can see a day that I can correspond with international people in English.

我能预见未来有一天, 我可以用英语与国际人士通信交流。

Appendix B. Writing prompts

1. Although technology has made communication easier, there are still as many misunderstandings among people as there were in the past.
What is your opinion?
Use specific reasons and examples to support your answer.
An effective essay is no less than 150 words.
2. Great changes have occurred in education. Please write an essay on the topic Changes in the Way of Education.
Use specific reasons and examples to support your answer.
An effective essay is no less than 150 words.
3. The relationship between parents and children is very important. Please write an essay on how to best handle the relationship between parents and children.
Use specific reasons and examples to support your answer.
An effective essay is no less than 150 words.
4. Smartphones have become an important part of people's life. Some say that we spend too much time on smartphones while others believe that more time should be spent on smartphones.

What is your opinion?

Use specific reasons and examples to support your answer.

An effective essay is no less than 150 words.

Appendix C. Rating criteria

13-15

The essay effectively addresses the topic. The ideas are expressed clearly, the text is coherent and mostly free of language errors, with only few minor language errors.

10-12

The essay effectively addresses the topic. The ideas are expressed clearly, the text is coherent, but there are a few language errors.

7-9

The essay basically addresses the topic. Some ideas are not expressed clearly, and the text is barely coherent. There are quite a few language errors, some of which are serious.

4-6

The essay basically addresses the topic. The ideas are unclear, lacking coherence, and there are numerous serious language errors.

1-3

The essay lacks organization. The ideas are chaotic. The language is fragmented and most sentences contain errors, with the majority containing serious errors.